

Innovation persistence: Survey and case-study evidence

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Abstract

Evidence on the persistence of innovation sheds light on the nature of the innovation process and can guide appropriate policy development. This paper examines innovation persistence in Ireland and Northern Ireland using complementary quantitative and case-study approaches. Panel data derived from innovation surveys is used, and suggests very different results to previous analyses of innovation persistence primarily based on patents data. Product and process innovation are found to exhibit strong general persistence but we find no evidence that persistence is stronger among highly active innovators. Our quantitative evidence is most strongly consistent with a process of cumulative accumulation at plant level. Our case-studies highlight a number of factors which can either interrupt or stimulate this process including market volatility, plants' organisational context and regulatory changes. Notably, however, the balance of influences on product and process innovation persistence differs, with product innovation persistence linked more strongly to strategic factors and process changes more often driven by market pressures.

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1. Introduction

Our understanding of the persistence of innovation is important in terms of our conceptual view of the underlying processes which drive innovation (e.g. the Schumpeter Mark I versus Mark II debate; [Malerba et al., 1997](#)). It is also important in strategic and policy terms, establishing whether investments (either private or public) in firms' innovation capability are likely to result

in organisational learning and subsequent innovation events ([Gruber, 1990](#); [Geroski et al., 1997](#)). Despite its importance, the existing knowledge-base on the persistence of firms' innovation activities is relatively narrow, and comes primarily from a series of studies which have adopted a quantitative approach to examining firms' patenting behaviour (e.g. [Malerba et al., 1997](#); [Geroski et al., 1997](#); [Malerba and Orsenigo, 1999](#); [Cefis and Orsenigo, 2001](#); [Cefis, 2003](#)), and innovation activity ([Geroski et al., 1997](#)). Overall, these studies find little evidence of widespread persistence of patent activity, but do identify a core of, generally larger, firms which patent persistently. [Geroski et al.](#)'s analysis of the SPRU database largely confirms this view, with innovation per-

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sistence depending strongly on high initial levels of innovation activity.¹ The group of firms which patent persistently, or in the case of Geroski et al. (1997) innovate persistently, are seen as important, however, both because they: generate a relatively large proportion of all patents (Geroski et al., 1997; Cefis and Orsenigo, 2001); and, because these firms show marked evidence of the type of creative accumulation envisaged in the Schumpeter Mark II model (Malerba and Orsenigo, 1999).²

The strength of this group of studies is in the insight they provide into the anatomy of innovation persistence; their main weakness, acknowledged by most authors, is their lack of explanatory power due primarily to the limited set of firm-level characteristics variables available in the patent and innovation datasets used (Geroski et al., 1997, p. 43).³ The studies by Cefis and Orsenigo (2001) and Cefis (2003), for example, reflect the sector and size of patenting firms but are unable to ‘capture the important dynamic dimension of strategy development, for example, the process of strategic adaptation and organisational learning’ and ‘soft’ behavioural issues (Borch and Arthur, 1995, p. 425). To try and overcome this issue we adopt here a mixed methods approach of the type advocated by Borch and Arthur (1995), combining a quantitative analysis of innovation persistence using innovation panel data with a series of case-studies designed to illustrate the behavioural and strategic determinants of innovation persistence.

Both elements of the analysis are based around the Irish Innovation Panel (IIP) which provides longitudinal data on plants’ innovation activities over the 12-year

period from 1991 to 2002 inclusive. This source has two major advantages over the data used in the majority of other studies of innovation persistence. First, like the SPRU database used by Geroski et al. (1997), the IIP provides direct rather than indirect evidence of the product and process innovations introduced by plants and the persistence of this innovation activity. Secondly, along with this innovation data, we have a wealth of detail on the plants themselves, and the way in which they approach and organise their innovation activities. This feeds into the case studies and our analysis of the behavioural and strategic practices which shape innovation persistence.

Our analysis does, of course, also inevitably reflect the specific factors which have shaped the innovation systems of Ireland and Northern Ireland. These we have described in detail elsewhere (i.e. Cooke et al., 2003; O’Malley et al., 2006), but it is worth recalling that during the period covered by our analysis Ireland, in particular, benefited from significant inward investment from the US (e.g. Roper and Frenkel, 2000). Concentrated in high-tech sectors, this investment tended to exacerbate differences between locally and externally owned plants, with externally owned plants tending to invest more in R&D, being more likely to be innovative, and tending to lead locally owned companies in the adoption of advanced manufacturing technologies (Roper and Hewitt-Dundas, 1998; Hewitt-Dundas et al., 2002; O’Malley and Roper, 2003).⁴ In addition to the sectoral differences in persistence highlighted by Malerba et al. (1997) and Cefis and Orsenigo (2001), our analysis will therefore also need to reflect potential differences in innovation persistence between locally owned and externally owned plants.

The remainder of the paper is organised as follows. Section 2 outlines our empirical approach, defining persistence and summarising our quantitative methodology which is based on the non-parametric approach advocated by Cefis and Orsenigo (2001) and Cefis (2003). Section 3 reports our results, integrating our quantitative and case-study analysis. Section 4 discusses the key empirical conclusions.

2. Data and measurement issues

Our persistence analysis is based around data from the Irish Innovation Panel (IIP) which provides information

¹ An essentially similar result is found by Kurokawa et al. (2002) in their study of the persistence of firm and university R&D into semiconductor laser diodes. They found that the more success organisations had in this research area (i.e. the more papers and patents the organisation had) the more likely they were to persist with their innovation efforts.

² In addition, these studies tend to identify international differences between patterns of patenting, and significant differences between sectors which are more pronounced than those between countries (Malerba et al., 1997; Cefis and Orsenigo, 2001). Cefis and Orsenigo (2001), for example, find consistent evidence across the six countries they consider that innovation persistence is strongest in chemicals followed by electrical and electronics machinery and mechanical engineering. For example, Cefis and Orsenigo (2001) find the following autoregressive parameters for chemicals (mechanical engineering) for a simple two state transition probability matrix (Tables 4 and 6): France 0.4299 (0.1992); Germany 0.4969 (0.3438); Italy 0.3856 (0.1987); Japan 0.6078 (0.5701) and the UK 0.5924 (0.2028).

³ Other potential critiques relate to the use of patent data as an indication of innovation (see the discussion in Malerba et al., 1997) and the failure to take into account the complexities of the corporate and economic contexts within which firms are operating.

⁴ This contrasts with the evidence of Geroski et al. (1997), p. 44, who note that in their analysis of UK firms ‘independent innovating units (marginally) outperform subsidiaries as persistent innovators, presumably because stronger investment incentives out-weigh any failure to exploit economies of scale’.

Table 1
Descriptive statistics

	1991–1993	1994–1996	1997–1999	2000–2002	1991–2002
	N = 877	N = 752	N = 1055	N = 920	N = 3604
A. Innovation indicators					
Product innovator (% plants)	64.2	66.1	63.9	58.3	63.0
Process innovator (% plants)	n.a.	59.0	63.1	54.9	59.2
Innovation intensity (% sales)	35.0	31.0	29.0	27.9	30.6
B. Sectoral composition					
Food, drink and tobacco	17.8	19.4	17.1	17.0	17.7
Textiles and clothing	14.0	12.4	11.1	7.5	11.3
Wood and related products	7.1	6.6	4.2	4.2	5.4
Paper and printing	6.6	7.2	6.7	8.2	7.2
Chemicals	7.4	8.1	6.2	5.8	6.8
Metals and metal fabrication	9.0	8.6	11.1	11.7	10.2
Mechanical engineering	8.2	7.9	7.6	7.9	7.9
Electrical and optical equipment	12.3	13.7	12.9	12.1	12.7
Transport equipment	4.6	3.7	3.5	2.7	3.6
Other manufacturing	13.0	12.4	19.5	22.9	17.2
C. Size mix					
Employment less than 50 (%)	46.0	45.2	52.9	59.5	51.5
50–249 employees (%)	42.5	41.4	37.6	32.3	38.1
250 or more employees (%)	11.5	13.4	9.5	8.3	10.4
D. Ownership					
Locally owned (%)	66.9	66.2	72.2	76.0	70.8
Externally owned (%)	33.1	33.8	27.8	24.0	29.2

Notes: a product innovator is a plant having introduced a new or improved product during the period covered by each survey (3 years). A process innovator is a plant having introduced a new or improved process over the same period. Innovation intensity is the proportion of sales derived from products which have either been newly introduced or improved over the previous 3 years.

on the innovation, technology adoption, networking and performance of manufacturing plants throughout Ireland and Northern Ireland over the period 1991–2002.⁵ The IIP comprises four linked surveys conducted using similar postal survey methodologies, sampling frames provided by the economic development agencies in Ireland and Northern Ireland, and questionnaires with common questions. Each survey covers the innovation activities of manufacturing plants with 10 or more employees over a 3-year period with an average response rate of 34.5%.⁶ The full panel consists of 3604 obser-

vations covering 2277 individual plants. Of these, 115 plants responded to all four waves of the IIP, 222 responded to three waves, 538 plants responded to two waves, and the remaining 1402 plants responded to only one of the four surveys.⁷ Table 1 provides an overview of the structure of the IIP focussing on the main variables used in our analysis. The tendency, as survey penetration and overall response has increased during the 3rd and 4th waves has been for more smaller and locally owned plants to be included in the group of respondents (Table 1). In terms of sector, the survey reflects the continuing importance of food manufacturing in Ireland and Northern Ireland and the declining importance of traditional sectors such

⁵ Our study provides the first analysis of innovation persistence in Ireland. Indeed, it could be argued that it constitutes the first analysis of innovation persistence in a small country as previous studies have tended to focus on patent data from larger countries. Cefis and Orsenigo (2001) adopt the broadest geographical canvas and conclude that 'Japan and Germany are typical Schumpeter Mark II (creative accumulation) countries, whilst Italy is a typical Schumpeter Mark I (creative destruction) country'. The USA, UK and France are in between these extremes (p. 1147).

⁶ Details of each wave of the survey can be found in Roper et al. (1996), Roper and Hewitt-Dundas (1998), Roper and Anderson (2000), Roper et al. (2004).

⁷ In large part this pattern reflects changes in sample structure and non-response but it may also reflect the inclusion in the sample of plants which were either newly opened or had closed at the other survey dates. Non-response analysis conducted with each survey (see, for example, Roper and Hewitt-Dundas, 1998), however, has suggested there is no systematic bias in the responses to each wave of the IIP which should therefore be representative of the target population of manufacturing plants.

as textiles and clothing and wood and related products.

In our quantitative analysis we focus on three main innovation indicators. First, whether plants have introduced any new or improved products over the 3-year period covered by the survey. Overall, around 63% of plants were in this category, although this proportion fell markedly in 2000–2002 due primarily to deteriorating economic conditions (see Roper et al., 2004 for discussion). Second, we consider whether plants introduced any new or improved processes during the previous 3 years. Overall, around 59% of plants in Ireland and Northern Ireland were in this category, although this proportion too fell during the 2000–2002 period. Third, we consider innovation intensity, i.e. the proportion of sales derived from products either newly introduced or improved over the last 3 years. On average, 31% of plants' sales were in this category although, reflecting lower levels of product innovation activity, this proportion too fell somewhat in the 2000–2002 period (Table 1).

The main advantage of the IIP for examining innovation persistence is the richness of the data available and the potential insights into the reasons for innovation persistence. Issues do arise, however, because of the nature of the IIP, i.e. a plant-level, relatively short, unbalanced panel with each observation covering innovation activity over a 3-year period. The first issue is that the IIP is a plant rather than firm-level survey, although it is possible to identify those business units which are part of wider groups (see, for example, Roper and Love, 2002).⁸ This reflects the structure of the business unit analysis of the SPRU database reported in (Geroski et al., 1997), but differs from the firm-level analysis which characterises those studies based on patents data (e.g. Cefis and Orsenigo, 2001). The implication is that our analysis – per force – is of the persistence of innovation at the level of the plant rather than of the firm. Of course, for single plant companies the two are identical, but for multi-plant groups this means our plant-level analysis is likely to suggest lower levels of innovation persistence than if we were working at firm level where the results might reflect the combined innovation activity of a number of group plants.

The second issue relates to the structure of the IIP which is a relatively short unbalanced panel. This

severely restricts our quantitative approach, effectively ruling out the hazard rate methodology used by Geroski et al. (1997), and suggesting instead the non-parametric approach adopted by Cefis and Orsenigo (2001). This focuses attention on the probability that a plant will continue to innovate from one period to the next, a notion expressed in the transition probability matrix (TPM). Suppose, for example, that I_t is a binary innovation indicator which takes value 1 if a plant is innovating, and zero otherwise. Then each term in the (2×2) TPM will be the conditional probability $p_{ij} = P(I_t = j | I_{t-1} = i)$, or the probability of moving from state i to state j . Of particular interest here are the probabilities on the leading diagonal of the TPM which represent the probability of remaining in the group of innovating or non-innovating plants. Following Cefis and Orsenigo (2001), we distinguish three scenarios for a 2×2 TPM⁹:

- (a) *Transient innovation*—where the sum of the leading diagonal terms of the TPM is less than one i.e. $\sum_i P(I_t = i | I_{t-1} = i) < 1$.
- (b) *Weak innovation persistence* where $\sum_i P(I_t = i | I_{t-1} = i) \geq 1$ but $P(I_t = i | I_{t-1} = i) < 0.5$ for either $i = 1$ or 2 .
- (c) *Strong innovation persistence* where $\sum_i P(I_t = i | I_{t-1} = i) \geq 1$ and $P(I_t = i | I_{t-1} = i) > 0.5$ for both $i = 1, 2$.

In the latter, most demanding, case we require that both innovators and non-innovators have a more than even chance of remaining in the same group during the next period. Such situations are likely to be relatively unusual, however, as Table 2 derived from Cefis and Orsenigo (2001) suggests. In their results for their whole sample, strong persistence is only observed in Japan, although weak persistence is ubiquitous, due primarily to persistence by firms in the non-innovating group.

Two approaches are possible to constructing TPMs from unbalanced panels such as the IIP. First, it is possible to extract only the 'balanced' element of the panel and use this data to construct the TPMs. In the case of the IIP this would mean focussing the analysis on the 115 plants which responded to all four waves of the survey, giving

⁸ The original reason for adopting this approach was concern over the ability of Irish business units to absorb new technology and effectively implement innovations (see for example, Roper and Thanki, 1995). The focus was therefore on the factors which were shaping the innovative performance of individual plants rather than the firm. See Love et al. (1996) for further discussion.

⁹ This is easily generalised to an $n \times n$ TPM with a requirement for strong persistence, for example, that $\sum_i^n P(I_t = i | I_{t-1} = i) > n/2$ and that $P(I_t = i | I_{t-1} = i) > 0.5$ for all $i = 1, n$.

Table 2
Persistence in innovation transition policy matrices

	Derived from:		
	1-year TPM	5-year TPM	10-year TPM
France	1.297	1.187	1.078
Germany	1.345	1.262	1.216
Italy	1.201	1.083	1.032
Japan	1.546*	1.489*	1.422*
UK	1.281	1.148	1.081
USA	1.290	1.204	1.101

Notes: Numbers are derived as the sum of the leading diagonal terms in the 2×2 TPMs reported by Cefis and Orsenigo. These relate to transition probabilities over a 1-, 5- and 10-year period regardless of firms' situation in intervening years. *Strong persistence, i.e. both the probability of remaining in the innovative and non-innovative group are greater than 0.5. Source: Cefis and Orsenigo (2001), Table 3.

us a working sample of 345 period-to-period transitions. The second potential approach is to construct a pooled sample of all period-to-period transitions in the database, i.e. all situations where plants responded to two consecutive waves of the panel. This gives us a sample of around 1210 period-to-period transitions.¹⁰ The main advantage of the first balanced panel approach is its simplicity. It does, however, mean throwing away nearly two-thirds of the observed transitions in the IIP and essentially focussing the analysis on what is more likely to be an atypical group of plants. Our preference therefore is to adopt the second approach and base our analysis on the largest possible set of period-to-period transitions.

In addition to binary innovation indicators for product and process innovation in the IIP, we also examine the relationship between innovation persistence and the intensity of plants' previous innovation activity. This follows Cefis and Orsenigo (2001) and Geroski et al. (1997) who suggest that within the group of innovators the probability of transition to the non-innovating group depends significantly on initial innovation intensity.¹¹ Here, we focus on the proportion of sales derived from innovative products, and classify plants into four categories: those with no sales of innovation products; those for whom sales of innovative products account for less than a third of turnover; those for whom sales of innovative products account for between a third and two-thirds of sales; and,

those for whom innovative sales accounts for more than two-thirds of total sales.

The final measurement issue raised by the IIP relates to the empirical meaning of the term persistence. In both the patenting studies of Cefis and Orsenigo (2001) and Geroski et al. (1997) the data used is annual, or at least is recorded annually. Persistence is therefore measured as the continuity of patenting (or innovation) from 1 year to the next. In the IIP, however, each wave of the panel covers plants' innovation activity over a 3-year period. This restricts our comparison to one 3-year period with another, a less demanding notion of persistence than that required by annual data. For example, in our analysis – as long as they innovate once in each 3-year period – plants' innovation would be classed as persistent, whereas they might fall into either the transitional groups or non-persistent groups if considered using annual data. This will tend to bias our results towards persistence, counteracting the negative effect on observed levels of persistence from our use of plant rather than firm-level data.

For the qualitative element of our analysis case studies were selected strategically to illustrate the factors underlying some of the more typical patterns of persistence. Our objective here was not to identify representative results but to adopt an inductive approach which might provide some insight into the complex of factors which shape innovation persistence in individual plants.¹² Ten businesses which had responded to all four waves of the IIP were initially identified as possible case studies on the basis of their patterns of innovation persistence, size and sector.¹³ A documentary and on-line search was then conducted for each plant to identify contextual material and any published case-study material.¹⁴ Interviews

¹⁰ Note that this is not the same as the overall number of repeated responses (i.e. situations in which plants responded to more than one survey) as some plants responded to non-consecutive waves of the IIP.

¹¹ Fritsch and Franke (2004) for example, working with German data, suggest that local spillovers have a strong positive effect on the probability of innovating but no significant effect on the probability that a firm will have high innovation intensity.

¹² In this approach we follow Verschuren (2003), p. 137: 'A case study is a research strategy that can be qualified as holistic in nature, following an iterative-parallel way of proceeding, looking only at a few strategically selected cases, observed in their natural context in an open ended way, explicitly avoiding (all variants) of tunnel vision, making use of analytical comparison of case or sub-cases, and aimed at description and explanation of complex and entangled group attributes, patterns, structures or processes'.

¹³ Our decision to focus on plants which responded to all four waves of the IIP was in order to have as long a history of innovation persistence as is possible within our data. As an anonymous referee pointed out, however, this might introduce a bias into the case studies due to these being surviving plants. In terms of our inductive objective here this is unlikely to be overly serious. It does suggest an interesting future research area however relating innovation persistence to firm or plant survival.

¹⁴ In one case, Clare Electronics, this review identified a relatively recent published case study which provided a detailed overview of the development of strategy and innovation in the company.

Table 3
Case study summary

	Case	Employment		Product innovation persistence	Process innovation persistence	Extent of product change
		50–100	100+			
1	Clare Electronics		✓	Persistent	Persistent	Stable, high
2	Claggan Design	✓		Persistent	Transitory	Increasing, now high
3	Toraigh Products		✓	Persistent	Transitory	Medium, variable
4	Rathlin Engineering		✓	Persistent	Transitory	Medium, stable
5	Dursey	✓		Transitory	Persistent	Variable, low
6	Foynes Cable		✓	Transitory	Persistent	Low variable
7	Inis Group		✓	Transitory	Transitory	Falling, now low
8	Islandmor Joinery Works	✓		Transitory	Transitory	Low stable

were then sought with the companies to explore in detail the influences on their innovation activity during the 1990s.¹⁵ Eight case-studies were ultimately completed, and an overview of the completed case-studies, their size and patterns of innovation persistence is given in Table 3 which also includes a brief overview of innovation intensity and its movement over the 1991–2002 period. Cases 1–4 were examples of plants which were persistent product innovators over the 1991–2002 period, but displayed different patterns of process innovation. Cases 5–8 were situations where product innovation was transitory but process innovation was either persistent or transitory. Broadly reflecting the balance in the IIP (Table 1), six of the cases were locally owned plants, and two (Cases 3 and 6) were externally owned.¹⁶

3. Empirical results

Across our entire sample we observe strong persistence for both product and process innovation in both single and two period transitions, i.e. both innovators and non-innovators are more likely to remain in these categories than change groups (Tables 4 and 6).¹⁷ In other words we observe overall levels of innovation persistence in Northern Ireland and Ireland which are higher than those observed by Cefis and Orsenigo (2001)

for non-Japanese countries. Here, for example, the sum of the leading diagonal terms of the TPM for product innovation persistence across a single period is 1.432, compared to single year parameters identified by Cefis and Orsenigo of 1.297 in France, 1.345 for Germany, 1.201 for Italy, 1.546 for Japan, 1.281 for the UK and 1.290 in Germany (Tables 2 and 4).

Perhaps more interesting, however, are contrasts between the patterns of persistence we observe for product and process innovation. Although both are strongly persistent, product innovation persistence is notably stronger than that for process innovation, with the difference being particularly marked over the longer term. In the two-period TPMs, for example, the sum of the leading diagonal terms for product innovation 1.400, but that for process innovation is 1.098. This is perhaps a surprising result given the evidence that product and process innovation decisions are strongly linked (e.g. Kraft, 1990). One possibility, however, is that the difference we observe reflects market conditions in Ireland during the 1990s, a period of particularly strong demand and profitability growth as the country established its Celtic Tiger reputation. Martinez-Ros (2000), for example, suggests that among Spanish firms expected profitability has a stronger (positive) effect on product innovation than process innovation, and that demand changes also have a stronger effect on the probability of product than of process change.¹⁸ If this is also the case among Irish firms, the difference we observe between product and process innovation persistence may well be an echo of the wider Celtic Tiger phenomena. An example of this type of innovation profile is provided by Case 7 (Inis Group Ltd.) where buoyant market demand encouraged prod-

¹⁵ Each interview was based on a feedback form to the company which summarised their responses to the IIP questionnaire over the previous 12 years. A semi-structured questionnaire was then used to explore further the balance of capability, resource and strategic influences on innovation. Copies of the interview schedule are available from the authors on request.

¹⁶ In the text the actual names of the companies have been changed.

¹⁷ We report non-parametric bootstrap errors for each TPM (see also Cefis and Espa, 1998). These were constructed using Limdep by repeated sampling (1000 times) with replacement from each sub-sample of data. The size of the repeat sample was determined by the actual sample size in each case. See Efron and Tibshirani (1993) for a general discussion.

¹⁸ In Martinez-Ros (2000) Table 5 for example which summarises the marginal effects the variable RECES which is a dummy variable taking value 1 in recession and 0 otherwise has coefficient –0.025 for product change and –0.020 for process change.

Table 4
Single period transition probability matrices: by size and ownership

Product innovation				Process innovation		
Entire sample	<i>n</i> = 1210	No	Yes	<i>n</i> = 698	No	Yes
	No	0.643 (0.023)	0.357 (0.023)	No	0.671 (0.028)	0.329 (0.028)
	Yes	0.211 (0.015)	0.789 (0.015)	Yes	0.325 (0.023)	0.675 (0.023)
Plant size	<i>n</i> = 566	No	Yes	<i>n</i> = 366	No	Yes
<50 employees	No	0.707 (0.027)	0.293 (0.027)	No	0.739 (0.032)	0.261 (0.032)
	Yes	0.297 (0.028)	0.703 (0.028)	Yes	0.396 (0.037)	0.604 (0.037)
Plant size	<i>n</i> = 482	No	Yes	<i>n</i> = 261	No	Yes
50–249 employees	No	0.564 (0.042)	0.436 (0.042)	No	0.582 (0.051)	0.418 (0.051)
	Yes	0.172 (0.020)	0.828 (0.020)	Yes	0.318 (0.036)	0.682 (0.036)
Plant size	<i>n</i> = 138	No	Yes	<i>n</i> = 70	No	Yes
250 plus employees	No	0.304 (0.097)	0.696 (0.097)	No	0.2 (0.129)	0.8 (0.129)
	Yes	0.113 (0.031)	0.887 (0.031)	Yes	0.133 (0.043)	0.867 (0.043)
Ownership	<i>n</i> = 686	No	Yes	<i>n</i> = 435	No	Yes
Indigenously owned	No	0.696 (0.027)	0.304 (0.027)	No	0.731 (0.030)	0.269 (0.030)
	Yes	0.269 (0.023)	0.731 (0.023)	Yes	0.392 (0.032)	0.608 (0.032)
Ownership	<i>n</i> = 329	No	Yes	<i>n</i> = 161	No	Yes
Externally owned	No	44.7 (0.054)	55.3 (0.054)	No	51.2 (0.076)	48.8 (0.076)
	Yes	14.3 (0.022)	85.7 (0.022)	Yes	26.3 (0.042)	73.7 (0.042)

Note: figures in parenthesis are non-parametric bootstrap errors. Source: Irish innovation panel.

uct innovation initially before a period of organisational and process change:

Case 7: Inis Group Ltd.

Inis Group Ltd. is a privately owned company established in 1979 which specialises in the creation of clean-room environments for the pharmaceutical, healthcare and optical equipment sectors. Inis have benefited substantially from inward investment to Ireland over the last decade, with employment growing from 60 in 1991 to around 300 currently. Product innovation in the firm was persistent during the 1991–1999 period as the company tried out new materials in its own manufacturing facility and sought to develop a standardised product range. A shift towards clean-room designs based on outsourced products post-1999 ended product innovation in the company. Process innovation has become more important, however, as the company has sought to develop IT systems to support its increasing emphasis on design and installation.

Case 3 (Toraigh Products) also provides an illustration of the impact of inward investment on innovation persistence in the context of the external acquisition of a locally owned firm and subsequent market re-orientation:

Case 3: Toraigh Products

Toraigh Products, originally established in 1959, traditionally served the UK and Irish markets with a range of high volume particle board products. Subject of an acquisition in the late 1980s, Toraigh was market re-oriented towards higher value added products. As a result, product innovation at Toraigh was persistent throughout the 1990s, although the contribution of these new products to sales has been limited due to limited marketing resources. Process changes at Toraigh have been more transitory, and by and large unrelated to the product changes which have been introduced. A major factor in the timing and extent of process innovation has been resource constraints in both the plant and group.

Alongside inward investment and high-tech growth, Ireland like other European economies has also experienced a contraction in some traditional sectors (Table 1). This market volatility too has provided a stimulus to innovation as in Case 5 (Durseley) and Case 8 (Islandmor Joinery Works Ltd.) although in both cases market contraction has led to only transitory product innovation:

Case 5: Dursey

Durseley is an independent, family-owned firm, founded in 1943 and currently employing around 60 people. The firm manufactures a range of flexible

Table 5
Single period transition probability matrices for product innovation intensity

Entire sample	<i>n</i> = 1179	None	Low	Med	High
	None	0.697 (0.021)	0.174 (0.022)	0.065 (0.027)	0.063 (0.031)
	Low	0.233 (0.017)	0.569 (0.027)	0.138 (0.034)	0.06 (0.031)
	Med.	0.173 (0.011)	0.33 (0.019)	0.372 (0.035)	0.126 (0.036)
	High	0.195 (0.011)	0.195 (0.013)	0.268 (0.024)	0.341 (0.037)
Plant size	<i>n</i> = 566	None	Low	Med	High
<50 employees	None	0.763 (0.025)	0.13 (0.036)	0.06 (0.049)	0.047 (0.060)
	Low	0.277 (0.019)	0.553 (0.042)	0.135 (0.054)	0.035 (0.061)
	Med.	0.239 (0.014)	0.268 (0.029)	0.38 (0.058)	0.113 (0.051)
	High	0.291 (0.012)	0.236 (0.016)	0.164 (0.037)	0.309 (0.061)
Plant size	<i>n</i> = 463	None	Low	Med	High
50–249 employees	None	0.606 (0.038)	0.263 (0.030)	0.066 (0.030)	0.066 (0.041)
	Low	0.227 (0.034)	0.552 (0.035)	0.141 (0.046)	0.08 (0.041)
	Med.	0.103 (0.020)	0.402 (0.024)	0.379 (0.047)	0.115 (0.048)
	High	0.184 (0.019)	0.197 (0.019)	0.289 (0.031)	0.329 (0.051)
Plant size	<i>n</i> = 138	None	Low	Med	High
250 plus employees	None	0.4 (0.101)	0.28 (0.049)	0.12 (0.070)	0.2 (0.045)
	Low	0.122 (0.094)	0.659 (0.071)	0.146 (0.079)	0.073 (0.062)
	Med.	0.188 (0.068)	0.281 (0.056)	0.344 (0.086)	0.188 (0.090)
	High	0.063 (0.080)	0.125 (0.043)	0.375 (0.072)	0.438 (0.094)
Ownership	<i>n</i> = 686	None	Low	Med	High
Indigenously owned	None	0.74 (0.024)	0.141 (0.032)	0.061 (0.043)	0.058 (0.050)
	Low	0.273 (0.019)	0.536 (0.037)	0.12 (0.047)	0.071 (0.050)
	Med.	0.217 (0.013)	0.261 (0.024)	0.391 (0.049)	0.13 (0.047)
	High	0.253 (0.013)	0.253 (0.020)	0.203 (0.034)	0.291 (0.052)
Ownership	<i>n</i> = 329	None	Low	Med	High
Externally owned	None	0.54 (0.055)	0.276 (0.036)	0.069 (0.045)	0.115 (0.037)
	Low	0.144 (0.048)	0.649 (0.047)	0.144 (0.058)	0.062 (0.045)
	Med.	0.159 (0.027)	0.362 (0.036)	0.333 (0.057)	0.145 (0.058)
	High	0.000 (0.035)	0.143 (0.024)	0.333 (0.043)	0.429 (0.061)

Note: figures in parenthesis are non-parametric bootstrap errors. Source: Irish innovation panel.

packaging films and distributes plastic, foil and paper packaging, in addition to catering and hygiene disposables. Declining demand from the firms' traditional markets and increasing labour cost pressures in the early 1990s forced the firm to restructure its product range to target a niche market characterised by more advanced manufacturing processes and higher value added. This restructuring involved a major reinvestment in new equipment and training and a short burst of product innovation activity. Process innovation at Dursey has been incremental throughout the 1990s with a focus on efficiency—improving yields, reducing energy costs and improving quality. Test equipment was also purchased to allow the firm to respond more effectively to customer requirements.

Case 8: Islandmor Joinery Works Ltd.

Islandmor Joinery Works Ltd. is a family-owned business, established in 1969 and currently employing

around 60 people. Historically, the firm specialised in the manufacture of domestic kitchens, fitted bedrooms and contract furniture largely for the local market. From 1991 to 1997 adequate margins from the firms' traditional markets discouraged product-led innovation, although process changes were introduced with the introduction of CAD and CNC machinery. Post-1998, in the face of declining demand in its traditional markets, the firm developed a new range of flat-pack products for export markets utilising its new equipment. The shift from serving local to export markets, and the associated increase in production volumes, created a need for significant changes in organisational behaviour and work practices within the firm.

Another observation made by Cefis and Orsenigo (2001), and others, is that of bimodality in the pattern of innovation persistence. In other words, persistence is strongest where firms are either non-innovators, or

are highly active innovators.¹⁹ In the case of France, for example in a single year TPM Cefis and Orsenigo (2001), Table 2, find that 89.0% of non-innovators remain in the same category and 74.5% of highly active innovators remain in the same group. By contrast, only 13.9 and 27.5% of innovators in the intermediate groups remain within the same groups. No such pattern of bimodality is evident in our data for the persistence of product innovation. Instead, we find a monotonic relationship with the probability of persistence being inversely related to the proportion of firms sales derived from innovative products (Tables 5 and 7). The implication is that in Ireland and Northern Ireland plants find it difficult to sustain high innovation intensity in the longer term. Indeed, for our whole sample only a third of plants were able to sustain a situation where more than two-thirds of sales were derived from innovative products from one (3-year) period to the next (Table 5).

It is important to acknowledge, however, that our results and those of Cefis and Orsenigo (2001) in this respect are not necessarily contradictory. The different periodicity, innovation indicators and measurement levels (i.e. plant rather than firm) are all likely to be contributing to the difference in results. For example, it is clear from other studies that the link between patent data, new product introductions and plants' sales is often weak (e.g. Hall and Bagchi-Sen, 2002). Moreover, if at plant level, investment in patenting and new products is lumpy or cyclical this would suggest the limited persistence of innovative sales we observe, but if aggregated to firm level could also generate the bimodality observed by Cefis and Orsenigo (2001). This type of aggregation effect would be exaggerated by the different periodicity of our analysis, based on a comparison of 3-year periods, and the annual data of Cefis and Orsenigo. Perhaps more interesting, however, is a conjecture suggested by the results of both studies, that while firms may be able to sustain active patenting it is less easy to maintain a consistently high level of commercially successful innovation, and generate a flow of sales from innovative products.

Given these general results on the persistence of innovation in Ireland and Northern Ireland, and following Martinez-Ros (2000) and other studies which have emphasised resource constraints on innovation in smaller firms, we might also expect to find stronger innovation persistence among larger firms. In fact, however, in terms of the single period TPMs (Tables 4 and 5),

we find strong innovation persistence among plants with less than 50 employees, and 50–249 employees and weak persistence among larger plants. Case 2 (Claggan Design Ltd.), for example, is a small independent company, which engaged in persistent product and process innovation throughout the 1991–1999 period and then, because a change of strategy, ceased any innovative activity.

Case 2: Claggan Design Ltd.

Claggan Design Ltd. is an independent company which designs and manufactures a range of support devices for children with physical disabilities. Employment rose rapidly during the 1990s in line with increasing sales. From start-up, and through a subsequent period of consolidation, the company engaged in persistent design-led product innovation and upgrading of its product range. Alongside design changes, changes in materials necessitated process innovation in manufacturing and design facilitating the development of more standardised components. The subsequent re-engagement of the founder of the firm after a period away pursuing other interests led to a pause in both product and process innovation as the company was restructured towards a strategy based on an expanded design capability and outsourced manufacturing.

By contrast, in Case 6 (Foynes Cable), a larger plant, product innovation was transitory and essentially driven by factors external to the plant essentially by-passing the resource-based arguments suggested by Martinez-Ros (2000):

Case 6: Foynes Cable

Foynes Cable employs around 120 people in the manufacture of low voltage cables for the construction and power industries. The plant is part of an international group but operates largely independently within a territorial marketing strategy. The firms' markets are characterised by price competition in what is essentially a mature product market for a standardised product. Product innovation at the plant during the 1990s was spasmodic; from 1991 to 1993 innovation was necessary to meet regulatory changes after the fire on the London underground; from 1996 to 1999 period, new types of cable manufacture were introduced. The timing of the second of these innovation episodes was dictated by a long awaited investment approval from group HQ. Process change at Foynes Cable was persistent throughout our sample period, driven by a need to reduce costs. This was not unrelated, however, to the plant's product

¹⁹ In Cefis and Orsenigo (2001) the group of highly active innovators is defined as those having six or more patents in a year.

Table 6

Two period transition probability matrices: by size and ownership

		Product innovation		Process innovation	
Entire sample	<i>n</i> = 451	No	Yes	<i>n</i> = 202	No
	No	0.621 (0.036)	0.379 (0.036)	No	0.531 (0.076)
	Yes	0.22 (0.026)	0.78 (0.026)	Yes	0.433 (0.075)
Plant size <50 employees	<i>n</i> = 211	No	Yes	<i>n</i> = 109	No
	No	0.682 (0.045)	0.318 (0.045)	No	0.615 (0.061)
	Yes	0.346 (0.049)	0.654 (0.049)	Yes	0.5 (0.075)
Plant size 50–249 employees	<i>n</i> = 191	No	Yes	<i>n</i> = 75	No
	No	0.532 (0.063)	0.468 (0.063)	No	0.387 (0.089)
	Yes	0.132 (0.029)	0.868 (0.029)	Yes	0.455 (0.073)
Plant size 250 plus employees	<i>n</i> = 49	No	Yes	<i>n</i> = 18	No
	No	0.4 (0.024)	0.6 (0.024)	No	0 (0.073)
	Yes	0.182 (0.058)	0.818 (0.058)	Yes	0.188 (0.095)
Ownership Indigenously owned	<i>n</i> = 267	No	Yes	<i>n</i> = 136	No
	No	0.696 (0.043)	0.304 (0.043)	No	0.579 (0.056)
	Yes	0.246 (0.038)	0.754 (0.038)	Yes	0.5 (0.067)
Ownership Externally owned	<i>n</i> = 112	No	Yes	<i>n</i> = 41	No
	No	0.333 (0.091)	0.667 (0.091)	No	0.286 (0.127)
	Yes	0.2 (0.044)	0.8 (0.044)	Yes	0.37 (0.094)

Note: figures in parenthesis are non-parametric bootstrap errors. Source: Irish innovation panel.

innovation activities. Indeed, successful process innovation by the plant was said to be the trigger for the group investment approval.

Looking more closely at the TPMs, however, suggests that the probability of staying within the innovating group for both products and processes actually does increase monotonically with plant size (Tables 4 and 6). This may reflect resource advantages, but may also reflect strategic factors as suggested by Case 4 (Rathlin Engineering) and Case 1 (Clare Electronics Ltd.) both of which are larger plants. In Case 4, persistent innovation was largely the result of acquisition and product transfers within the group, while in Case 1, persistent innovation was more strongly linked to the plants' own strategic ambitions:

Case 4: Rathlin Engineering

Rathlin engineering was formed in 1979 and is part of an Irish consumer electronics group. The firm employs around 480 people, up around a quarter on 2000. Product innovation at Rathlin has been continuous, the result of transfers of products from other group plants to the firm as well as developments from the firms' own R&D department. Process innovation has been more transitory, driven largely by the needs of the plant's changing product portfolio.

Case 1: Clare Electronics Ltd.

An independent company, Clare Electronics was established in 1982 to design and manufacture electronic components for the automotive industry. The current workforce of 190 and dramatic sales growth throughout the 1990s is the result of persistent product and process innovation. Both have been closely aligned in achieving the strategic aims of the firm. For example, the introduction of improved and new products have been enabled by process changes in the firm, in the form of quality accreditation, environmental certification, the introduction of e-business and enterprise resource planning processes. Of particular importance in the innovation process at Clare have been the formation and development of external innovation links with technology partners, local universities research centres and industry research laboratories.

As suggested earlier, for Ireland and Northern Ireland is it also of considerable interest to examine differentials in innovation persistence between locally owned plants and those which are externally owned, although in some cases sample sizes are relatively small.²⁰ In partic-

²⁰ In our database some of the sectoral TPMs depend on relatively small samples and so detailed results are not reported. In general, however, we observe strong product innovation persistence in all sectors with the exceptions of paper and printing and electrical and optical equipment. Process innovation also proves strongly persistent in all

Table 7
Two period transition probability matrices for product innovation intensity

Entire sample	<i>n</i> = 339	None	Low	Med	High
	None	0.647 (0.037)	0.194 (0.050)	0.112 (0.047)	0.047 (0.044)
	Low	0.315 (0.030)	0.438 (0.053)	0.124 (0.076)	0.124 (0.041)
	Med.	0.128 (0.025)	0.404 (0.036)	0.277 (0.066)	0.191 (0.045)
	High	0.226 (0.016)	0.271 (0.034)	0.165 (0.057)	0.15 (0.049)
Plant size	<i>n</i> = 191	None	Low	Med	High
<50 employees	None	0.714 (0.045)	0.171 (0.090)	0.105 (0.110)	0.01 (0.077)
	Low	0.5 (0.037)	0.353 (0.088)	0.059 (0.108)	0.088 (0.055)
	Med.	0.294 (0.031)	0.235 (0.040)	0.412 (0.121)	0.059 (0.075)
	High	0.286 (0.009)	0.114 (0.051)	0.257 (0.059)	0.343 (0.081)
Plant size	<i>n</i> = 167	None	Low	Med	High
50–249 employees	None	0.55 (0.064)	0.233 (0.056)	0.133 (0.047)	0.083 (0.059)
	Low	0.196 (0.056)	0.5 (0.072)	0.196 (0.108)	0.109 (0.062)
	Med.	0.048 (0.045)	0.476 (0.058)	0.19 (0.089)	0.286 (0.075)
	High	0.15 (0.034)	0.175 (0.045)	0.325 (0.102)	0.35 (0.075)
Plant size	<i>n</i> = 41	None	Low	Med	High
250 plus employees	None	0.4 (0.239)	0.2 (0.144)	0 (0.019)	0.4 (0.111)
	Low	0.222 (0.195)	0.444 (0.175)	0 (0.169)	0.333 (0.120)
	Med.	0 (0.000)	0.556 (0.017)	0.222 (0.144)	0.222 (0.059)
	High	0.278 (0.238)	0.333 (0.168)	0.056 (0.144)	0.317 (0.110)
Ownership	<i>n</i> = 241	None	Low	Med	High
Indigenously owned	None	0.71 (0.041)	0.145 (0.072)	0.113 (0.076)	0.032 (0.064)
	Low	0.348 (0.032)	0.435 (0.074)	0.087 (0.096)	0.13 (0.053)
	Med.	0.16 (0.028)	0.36 (0.043)	0.28 (0.088)	0.2 (0.067)
	High	0.239 (0.016)	0.13 (0.049)	0.239 (0.080)	0.391 (0.074)
Ownership	<i>n</i> = 95	None	Low	Med	High
Externally owned	None	0.385 (0.097)	0.346 (0.092)	0.154 (0.000)	0.115 (0.075)
	Low	0.292 (0.095)	0.417 (0.096)	0.167 (0.153)	0.125 (0.079)
	Med.	0 (0.074)	0.5 (0.078)	0.25 (0.130)	0.25 (0.069)
	High	0.242 (0.063)	0.273 (0.066)	0.182 (0.125)	0.303 (0.079)

Note: figures in parenthesis are non-parametric bootstrap errors. Source: Irish innovation panel.

ular, we observe strong persistence of both product and process innovation among domestically owned plants in both the one and two period TPMs, but only weak innovation persistence for externally owned plants for product innovation and process innovation in the two period TPM (Tables 4 and 6). The implication is that in Ireland and Northern Ireland, as Geroski et al. (1997) found for the UK, innovation persistence in terms of both product and process is higher in locally rather than externally owned plants. Notably, however, for both product and process innovation, differences in persistence between locally

and externally owned firms are driven primarily by the higher probability that domestically owned firms will remain in the group of non-innovators.

Overall, our quantitative analysis suggests strong innovation persistence across most plant sizebands and ownership groups in Ireland and Northern Ireland, suggesting the importance of cumulative accumulation processes at the level of the individual plant. As our case studies suggest, however, this process of cumulative accumulation is strongly moderated by the market situation of the business (e.g. Cases 5 and 6), its organisational context (e.g. Cases 4 and 6) and its strategic aspirations (Cases 1 and 2). Regulatory effects may also influence innovation persistence (e.g. Case 6), while ownership changes may have either a positive (e.g. Case 4) or restraining (Case 6) effect on innovation activity. Technical change too, or more accurately technology adoption, may create new market opportunities and stimulate further innovation (e.g. Case 8).

sectors with the exception of chemicals. We also find no evidence at sectoral level of the type of bimodality of innovation persistence observed in patent based studies (e.g. Malerba et al., 1997; Geroski et al., 1997; Cefis and Orsenigo, 2001; Cefis, 2003). Instead, as before, we find that in all sectors persistence weakens monotonically as the proportion of innovative sales increases. Detailed sectoral tables are available from the authors on request.

Our quantitative results also differ from those of earlier patent-based studies of innovation persistence in that we find no evidence of ‘bimodality’ in innovation persistence. In particular, we find no evidence of higher levels of innovation persistence among plants which are more active innovators. Plants, it seems, are able to sustain persistent innovation but find it much more difficult to sustain high levels of sales of innovation intensity. This may simply reflect the uncertainty attached to most innovation activity, but may also be linked to investment cycles or the lifetime of specific new products. In Case 2, for example, prior to the re-engagement of the founder of the firm, innovation intensity fell markedly. Similarly, in Case 3 although product innovation had been taking place, limited marketing resources meant that this innovation had contributed little to innovation intensity (Table 7).

4. Conclusions

Previous quantitative analyses of innovation persistence, predominantly using patents data, have suggested that, in general, innovation persistence is limited, sectoral variations are substantial, and that a degree of bi-modality exists with the strongest persistence evident among the most active innovators. Based on innovation survey data, rather than patents data our quantitative analysis suggests somewhat different results. First, in our dataset covering Ireland and Northern Ireland, both product and process innovation are strongly persistent. In other words, from one 3-year period to the next, innovating plants are more than likely to continue innovating, while non-innovating plants are more than likely to remain non-innovators. Second, we are unable to find any evidence of bi-modality. Instead, we find that the persistence of high levels of sales of innovative products declines monotonically from its initial level. In other words, plants with high levels of sales of innovative products in one 3-year period find it hard to sustain this position through the next 3 years. This may reflect market pressures or, as our case studies suggest, more strategic issues related to firms’ innovation investment decisions.

Third, larger plants prove more able in sustaining product and process innovation than smaller plants reflecting perhaps either resource constraints in smaller plants (e.g. Martinez-Ros, 2000), different strategic priorities between plant sizebands or the cumulative accumulation of technological advantage in larger plants. Fourth, although some of our sectoral samples are relatively small, we observe broadly similar levels of innovation between sectors rather than the marked

sectoral variation in persistence found in patent-based analyses (e.g. Cefis and Orsenigo, 2001). In some ways, this contrast is unsurprising as previous studies have suggested that in many sectors patent activity and firms’ new product and process introductions are only weakly linked (e.g. Hall and Bagchi-Sen, 2002), and that the relationship between patents, other knowledge inputs, and innovation outputs is likely to differ between sectors (e.g. Love and Roper, 2001). Moreover, survey based analyses of innovation propensity etc. generally suggest more even levels of innovative activity (measured, for example, by the proportion of innovating firms) across sectors than those based on patents (e.g. Roper and Hewitt-Dundas, 2006). In methodological terms, however, the difference between our results and those based on patents does highlight the sensitivity of any analysis of innovation persistence to the use of different innovation indicators.

A more unusual element of our analysis is our examination of innovation persistence in indigenously and externally owned plants (although see also Geroski et al., 1997). For Ireland and Northern Ireland, where inward investment has been a significant feature of industrial development and industrial policy over the last decade, this distinction is, of course, of particular interest. The results have wider implications, however, in terms of the potential ability of foreign direct investment to generate sustained technology transfers and productivity spillovers to their host economy. Here, our results are reassuring in that externally owned firms prove more likely to sustain their product and process innovation activity than locally owned firms. In policy terms this suggests that the spillover benefits from public support for innovation projects in externally owned firms will be longer lasting than similar projects supported in indigenously owned plants, and therefore that the social returns from supporting innovation in externally owned plants might be strongest *ceteris paribus*.

More generally, the relatively high level of innovation persistence we observe in our quantitative analysis is strongly suggestive of a Schumpeterian Mark II innovation process, i.e. one based on the cumulative accumulation of technological capability at the level of the plant. Our case studies serve to illustrate some of the plant level factors which can either interrupt or facilitate this process, however. Market pressures, plants’ organisational context and strategic aspirations may all either encourage or interrupt persistent innovation, with similar effects evident from regulatory changes, changes in ownership and new technology adoption. One more specific observation is the differential effects of plants resource base, strategy, etc. on product innovation persistence on

the one hand, and process innovation persistence on the other (see also [Martinez-Ros, 2000](#)). In the majority of cases, product innovation persistence was most strongly linked to the strategy of the company, and its aspiration to maintain or change its market position. Process innovation, on the other hand, was more commonly associated with market structure and, in particular, the need to compete effectively in increasingly cost-sensitive markets. These different influences may reflect plants search for coherence, i.e. a strategic approach in which internal capabilities are matched to market needs (e.g. [Lissack and Roos, 2001](#)). Or, in an era of network-based competition and ‘open’ innovation, may reflect the breakdown of any very strong link between firms’ internal competencies and innovation outputs (e.g. [Chesborough, 2003](#)). In either case, a narrow capability-based explanation of innovation persistence is clearly inadequate and a more strategically aware, systemic and contextual approach is necessary (see, for example, [Iansiti and Levien, 2004](#)).

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.respol.2007.10.005](https://doi.org/10.1016/j.respol.2007.10.005).

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